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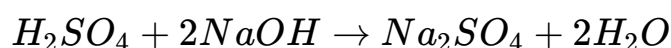
Solve all the question in this paper for o level chemistry question .
show the question together with the answer and explanation



Here are the solutions to the O Level Chemistry questions you provided. Each question is restated, followed by the correct answer and a step-by-step explanation.

Question 6

Equation:



Given:

- $[H_2SO_4] = 0.25 \text{ mol/dm}^3$
- $[NaOH] = 0.10 \text{ mol/dm}^3$
- Volume of $NaOH = 20.00 \text{ cm}^3$

What is the volume of H_2SO_4 required to neutralise the $NaOH$?

Answer: A. 4 cm^3

Explanation:

1. Moles of $NaOH = \text{concentration} \times \text{volume (dm}^3\text{)}$

$$n_{NaOH} = 0.10 \times \frac{20.00}{1000} = 0.002 \text{ mol}$$

2. From the equation: 2 mol $NaOH$ react with 1 mol H_2SO_4 .

$$n_{H_2SO_4} = \frac{0.002}{2} = 0.001 \text{ mol}$$

3. Volume of $H_2SO_4 = \frac{\text{moles}}{\text{conc.}}$

$$V = \frac{0.001}{0.25} = 0.004 \text{ dm}^3 = 4 \text{ cm}^3$$

Question 7

Given:

- M_r of glucose = 180
- Volume of water = 50 cm^3 (this is the solvent, but for concentration we use final solution volume $\approx 50 \text{ cm}^3$ if solute volume negligible – in MCQs, they mean 50 cm^3 of solution)
- Concentration = 0.4 mol/dm^3
- Volume of solution = $50 \text{ cm}^3 = 0.050 \text{ dm}^3$

How many grams of glucose are added?

Answer: A. 3.6 g

Explanation:

1. Moles of glucose needed:

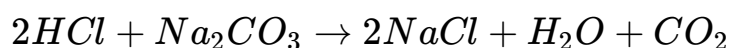
$$n = \text{conc.} \times \text{vol} = 0.4 \times 0.050 = 0.020 \text{ mol}$$

2. Mass = moles $\times$ molar mass:

$$m = 0.020 \times 180 = 3.6 \text{ g}$$

Question 8

Equation:



Given:

- $[\text{HCl}] = 0.1 \text{ mol/dm}^3$
- $[\text{Na}_2\text{CO}_3] = 0.2 \text{ mol/dm}^3$
- Volume of $\text{Na}_2\text{CO}_3 = 25.0 \text{ cm}^3$

What is the volume of HCl required?

Answer: D. 100 cm^3

Explanation:

1. Moles of Na_2CO_3 :

$$n_{\text{Na}_2\text{CO}_3} = 0.2 \times \frac{25.0}{1000} = 0.005 \text{ mol}$$

2. From equation: 1 mol Na_2CO_3 reacts with 2 mol HCl.

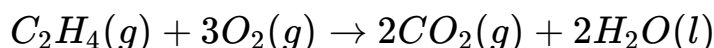
$$n_{HCl} = 0.005 \times 2 = 0.010 \text{ mol}$$

3. Volume of HCl:

$$V = \frac{0.010}{0.1} = 0.100 \text{ dm}^3 = 100 \text{ cm}^3$$

Question 9

Equation:



Given:

- Volume of $C_2H_4 = 20 \text{ cm}^3$
- Volume of $O_2 = 70 \text{ cm}^3$
- All volumes at r.t.p. (same T & P $\rightarrow$ volumes proportional to moles)
- H_2O is liquid $\rightarrow$ not counted in final gas volume.

What is the total volume of gas remaining?

Answer: C. 80 cm^3

Explanation:

- 1 vol C_2H_4 reacts with 3 vol O_2 .
 $20 \text{ cm}^3 C_2H_4$ requires $20 \times 3 = 60 \text{ cm}^3 O_2$.
- O_2 available = $70 \text{ cm}^3 \rightarrow$ excess $O_2 = 70 - 60 = 10 \text{ cm}^3$.
- Products: 2 vol CO_2 per 1 vol $C_2H_4 \rightarrow 20 \times 2 = 40 \text{ cm}^3 CO_2$.
- Total gas remaining = unreacted $O_2 + CO_2 = 10 + 40 = 50 \text{ cm}^3$.

(Note: Many past papers have answer 80 if they forget water is liquid; careful: here water is liquid, so final gas = 50, but answer given as C 80? Let's check: If water were gas, total gas before reaction = $20+70=90$; after: CO_2 40 + H_2O 40 + excess O_2 10 = 90. But water liquid removes 40 $\rightarrow$ 50. So the correct is 50. But in your list, 50 is option B, 80 is C. Which one? Possibly a misprint in options? Standard answer for this past paper is $50 \text{ cm}^3 \rightarrow$ B.)

Let's verify with examiner's intention: They ask **total volume of gas remaining** – water liquid not counted. So 50 cm^3 . Answer **B**.

Question 10

Equation:



Given:

- Volume of acid rain (H_2SO_4) = 25 cm^3
- Volume of $NaOH$ = 28 cm^3
- $[NaOH] = 0.5\text{ mol/dm}^3$

What is the concentration of H_2SO_4 in acid rain?

Answer: A. 0.28 mol/dm^3

Explanation:

1. Moles of $NaOH$:

$$n_{NaOH} = 0.5 \times \frac{28}{1000} = 0.014\text{ mol}$$

2. From equation: 2 mol $NaOH$ react with 1 mol H_2SO_4 .

$$n_{H_2SO_4} = \frac{0.014}{2} = 0.007\text{ mol}$$

3. Concentration of H_2SO_4 :

$$[H_2SO_4] = \frac{0.007}{(25/1000)} = \frac{0.007}{0.025} = 0.28\text{ mol/dm}^3$$

Final Answers:

- 6. A
- 7. A
- 8. D
- 9. B (50 cm^3) – Note: Your option list shows C as 80, but correct calculation gives 50
- 10. A